Topic 02: SIEM vs SOAR vs EDR — Understanding the Differences

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1. Introduction

When studying security operations, beginners often encounter three critical acronyms: SIEM, SOAR, and EDR. While all three are essential pillars of a modern Security Operations Center (SOC), they serve completely different functions. Confusing them can lead to architecture gaps and inefficient incident response. As an aspiring SOC analyst, understanding the distinct roles, core data sources, and operational workflows of SIEM, SOAR, and EDR is critical for investigating threats effectively.

2. What is SIEM?

Security Information and Event Management (SIEM) focuses primarily on centralized log collection, normalization, and real-time event correlation across the entire IT enterprise. A SIEM acts as the master aggregator, pulling telemetry from firewalls, domain controllers, cloud infrastructure, and endpoints into a single searchable data lake. Platforms like Splunk allow analysts to write queries to detect anomalous behaviors, meet compliance requirements, and maintain historical audit trails.

3. What is SOAR?

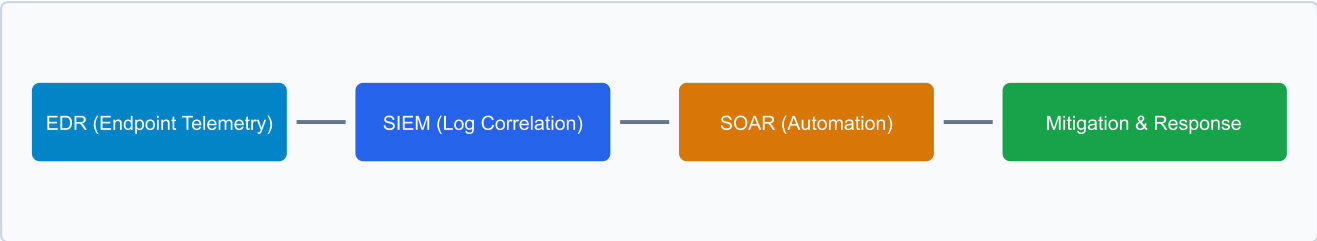
Security Orchestration, Automation, and Response (SOAR) focuses on streamlining and accelerating incident response through automated playbooks and tool integration. While a SIEM detects and flags a threat, a SOAR platform takes action by orchestrating tasks across different security solutions without manual intervention. For example, when a malicious IP is identified, a SOAR playbook can automatically query threat intelligence APIs, block the IP on perimeter firewalls, and notify the incident response team.

4. What is EDR?

Endpoint Detection and Response (EDR) focuses specifically on deep visibility and threat mitigation at the endpoint level (laptops, workstations, and servers). Unlike a SIEM that looks broadly across all network logs, EDR agents continuously monitor process executions, file modifications, memory injections, and registry changes. When malicious activity occurs on an endpoint, EDR provides deep forensic details and allows analysts to isolate the host from the network instantly.

5. How SIEM, SOAR, and EDR Work Together

The synergy between these three technologies creates a powerful defensive pipeline within a mature SOC environment:



6. Key Differences Between SIEM, SOAR, and EDR

To prevent operational confusion, analysts must remember the primary focus of each technology:

- **SIEM:** Focuses on *breadth*. It ingests logs from firewalls, active directory, cloud apps, and endpoints to correlate enterprise-wide security events.
- **EDR:** Focuses on *depth* at the endpoint. It provides granular process tracking, memory inspection, and host isolation capabilities that a SIEM cannot perform natively.
- **SOAR:** Focuses on *action and workflow*. It connects tools together through APIs to automate repetitive incident response steps and accelerate triage.

7. Comprehensive Comparison Table

Feature	SIEM	SOAR	EDR
Primary Purpose	Centralized log collection & event correlation.	Workflow automation & incident orchestration.	Endpoint threat detection & host containment.
Core Data Sources	Enterprise logs, firewalls, AD, cloud, EDR.	SIEM alerts, threat intel feeds, ticket systems.	Workstation/Server telemetry, kernel hooks.
Detection Mechanism	SPL/SQL queries, correlation rules, UEBA.	Triggered by external alerts (does not detect raw).	Behavioral analytics, file hashes, Sigma rules.
Investigation Depth	Broad enterprise log review across systems.	Gathers context automatically via API integrations.	Deep forensic process trees, file history, memory.
Response & Action	Generates alerts; limited direct remediation.	Automated playbooks across disparate tools.	Host isolation, process termination, file quarantine.
Automation Level	Rule-based alerting and scheduled searches.	Advanced end-to-end multi-tool workflow automation.	Automated blocking of known malware execution.

8. How These Three Technologies Work Together in a SOC

In an active SOC environment, these systems operate in a continuous feedback loop. An **EDR** agent detects a suspicious PowerShell script executing on a workstation and flags it. The EDR forwards its logs to the **SIEM**, where a correlation rule matches the endpoint event with an abnormal inbound VPN connection from an overseas IP. The SIEM fires an alert. A **SOAR** platform instantly intercepts this alert, runs an automated playbook to enrich the threat indicator against VirusTotal, isolates the compromised host via the EDR API, and opens an incident ticket for the analyst.

9. Practical SOC Scenario Demonstrating All Three Technologies

> **Scenario: Investigating a Mimikatz Credential Dumping Attack**

1. **EDR Detection:** The EDR agent detects memory dumping behavior associated with `mimikatz.exe` on workstation `WS-042`.
2. **SIEM Correlation:** The EDR forwards the event to Splunk. A correlation rule correlates the EDR alert with Windows Security Event ID 4688 (Process Creation) showing obfuscated command-line execution.
3. **SOAR Automation:** The SIEM triggers an incident in the SOAR platform. The SOAR executes a playbook that automatically isolates `WS-042`, extracts the user account, and triggers an urgent password reset in Active Directory.

10. Real-World Use Cases

- **Compliance Auditing (SIEM):** Storing financial transaction logs for PCI-DSS compliance audits.
- **Phishing Triage (SOAR):** Automatically parsing reported phishing emails, extracting URLs, checking reputation APIs, and deleting emails from user inboxes.
- **Ransomware Containment (EDR):** Automatically terminating encryption processes and rolling back affected files upon detecting ransomware signatures.

11. Benefits and Limitations of Each Technology

- **SIEM:** Great for visibility and compliance, but generates high alert volume and requires heavy tuning.
- **SOAR:** Eliminates repetitive manual tasks and speeds up response, but requires complex playbook maintenance.
- **EDR:** Provides unmatched endpoint forensic depth, but is blind to non-endpoint network traffic (like DNS or perimeter firewall activity).

12. Best Practices for SOC Analysts

- **Cross-Verify Data:** Never rely on a single tool; correlate SIEM logs with EDR process telemetry during investigations.
- **Test Playbooks:** Regularly audit SOAR automation playbooks to ensure API integrations do not fail during an active incident.

13. Conclusion

Studying SIEM, SOAR, and EDR side-by-side has clarified how modern security tools complement each other. While the SIEM provides enterprise-wide visibility, the EDR delivers deep endpoint forensics, and the SOAR automates rapid response actions. Mastering how these three platforms interact is essential for my growth as an effective SOC analyst.